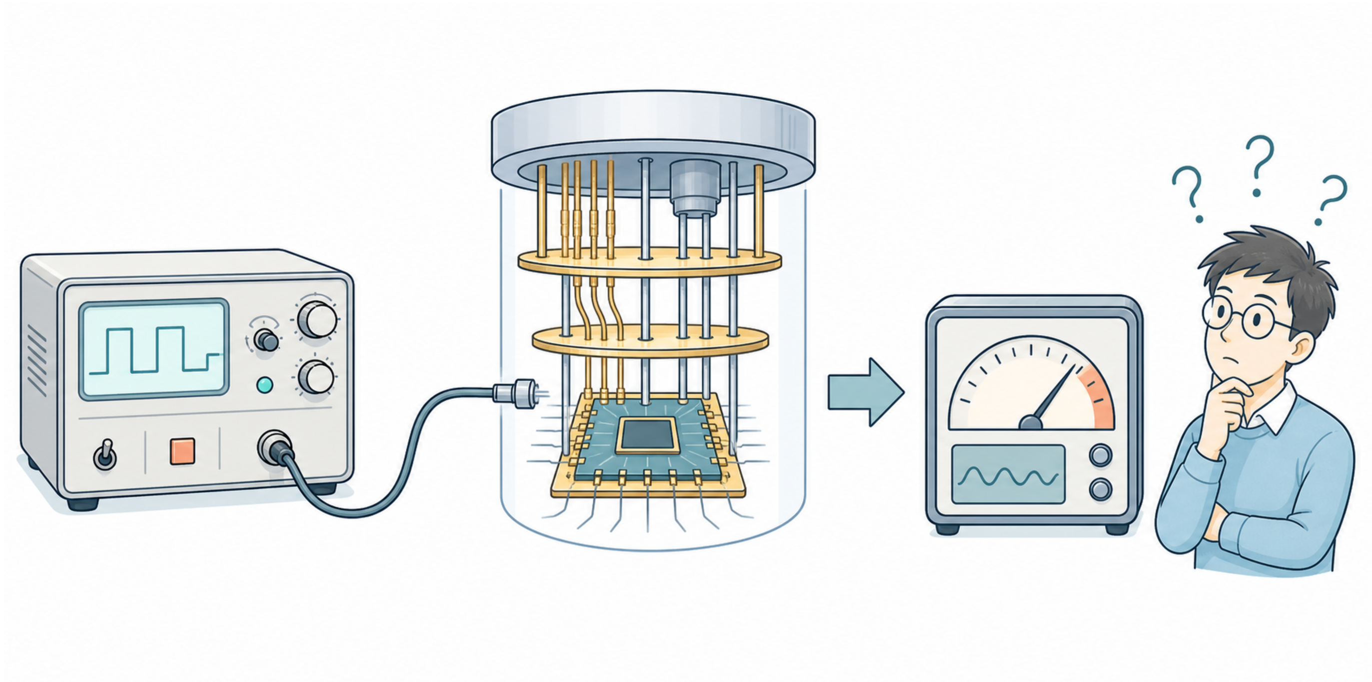
Universally Robust Control for a Qubit under an Unknown Perturbation

Mohammed Bilal P. S. Camille L. Latune Dominique Sugny

Université Bourgogne Franche-Comté, CNRS, Laboratoire Interdisciplinaire Carnot de Bourgogne (ICB) UMR 6303, F-21000 Dijon, France

Why Robust Control?



- Can a protocol remain resilient to experimental perturbation and uncertainty?
- How should such a robustness requirement be formulated?
- What cost may be paid to achieve it?

Mathematical Framework

We consider a controlled qubit evolving under a nominal control Hamiltonian $H_0(t)$ and a small unknown static perturbation αV:

$$H(t) = H_0(t) + \alpha V$$

The control field is written in the Pauli basis as

$$H_0(t) = \tfrac{1}{2}\big[u_x(t)\sigma_x + u_y(t)\sigma_y + u_z(t)\sigma_z\big],$$

and the perturbation is expanded in the same basis:

$$V = v_x\sigma_x + v_y\sigma_y + v_z\sigma_z.$$

Expanding the dynamics in powers of the perturbation strength α using the Dyson series, the final fidelity can be written in terms of the unperturbed states $|\psi_0(t)\rangle$ and $|\psi_\perp(t)\rangle$ as [1]:

$$|\langle\psi_{\text{target}} \mid \psi(t_f)\rangle|^2 = 1 - \alpha^2 \left| \int_0^{t_f} \langle\psi_\perp(t)|V|\psi_0(t)\rangle\, dt \right|^2 + \mathcal{O}(\alpha^4)$$

Here $|\psi_0(t)\rangle$ is the nominal trajectory state and $|\psi_\perp(t)\rangle$ is the state orthogonal to it.

State-to-State

From the decomposition of V in the Pauli basis we define:

$$F_k = \int_0^{t_f} \langle\psi_\perp(t)|\sigma_k|\psi_0(t)\rangle\, dt, \qquad k \in \{x, y, z\}.$$

The condition for second-order universal robustness is:

$$F_x = F_y = F_z = 0.$$

Optimization Strategy

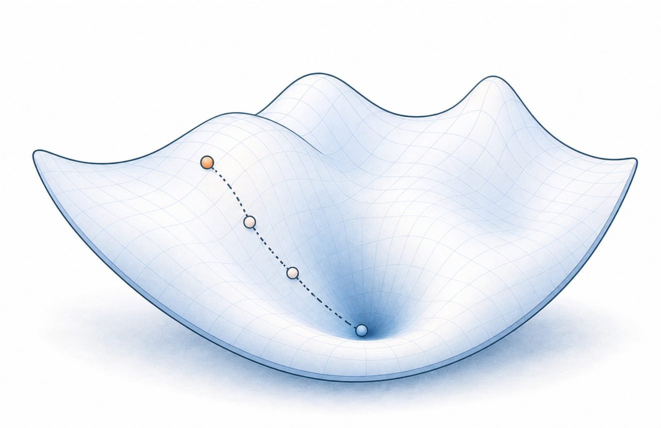
We optimize the control pulse to satisfy accuracy and robustness requirements. We express the cost function J as:

$$J = 1 - F(t_f) + \phi(t_f)$$

$$1 - F(t_f)\ :\ \text{target accuracy} \qquad \phi(t_f)\ :\ \text{robustness constraints}$$

For state-to-state transfer we have, $\phi(t_f) = \sum_k F_k^2$. The control is then updated along the gradient descent direction in an iterative loop with $\eta > 0$ as the step size until it converges to the optimal solution:

$$u^{(n+1)}(t) = u^{(n)}(t) - \eta\,\nabla J$$



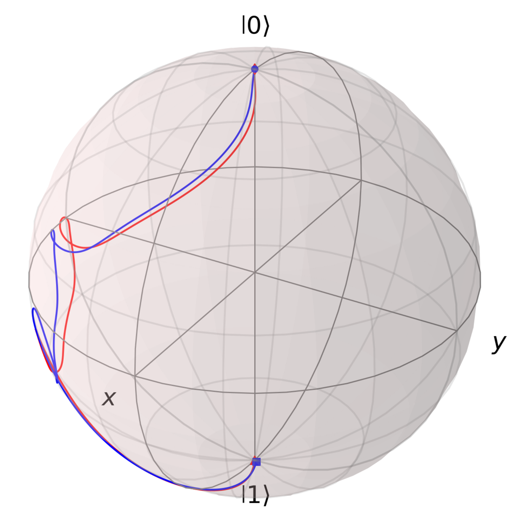
State-to-State Transfer

We seek a complete transfer from $|\psi_{\text{initial}}\rangle$ to $|\psi_{\text{target}}\rangle$ that remains robust under perturbations.

Key outcomes

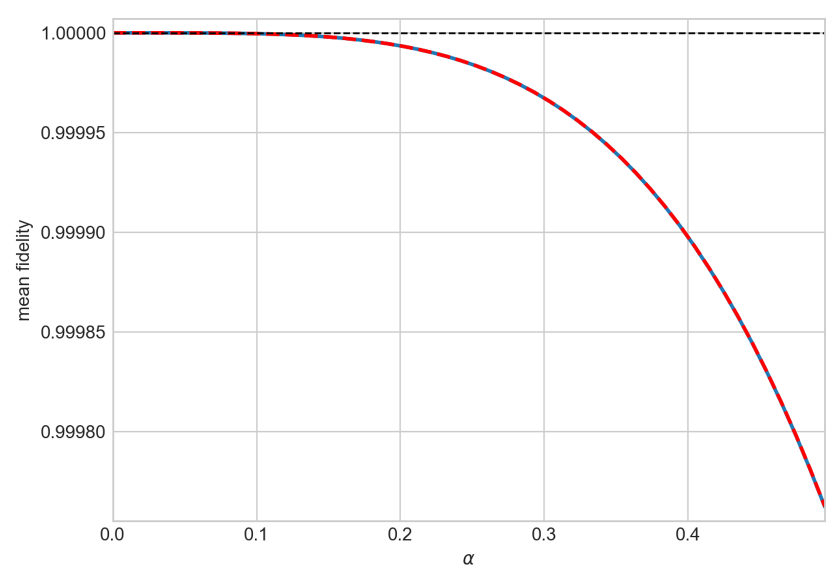
- The benchmark transfer $|0\rangle \to |1\rangle$ reaches nominal fidelity $\mathcal{F}_0 = 1$.
- The sphere-averaged fidelity remains near unity for small α and reaches $\langle\mathcal{F}\rangle_S \approx 0.9963$ at $\alpha = 1$ (well beyond the perturbative regime).
- Five representative state pairs, spanning pole, equatorial, and off-axis cases, retain $\langle\mathcal{F}\rangle_S > 0.992$ on $\alpha \in [0, 1]$.

$\mathcal{F}_0$: fidelity without perturbation ($\alpha = 0$)
$\langle\mathcal{F}\rangle_S$: fidelity averaged over 250 perturbation directions on the unit sphere.



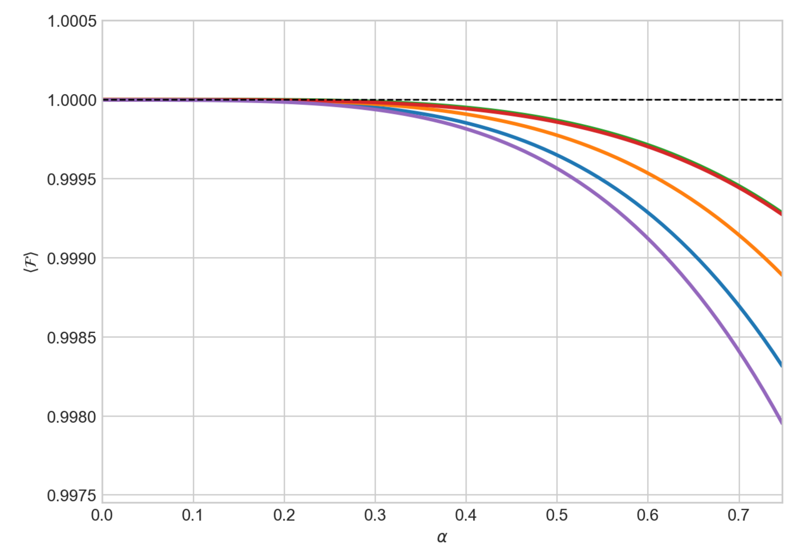
Bloch trajectories at $\alpha = 0.5$ remain close for $\alpha = 0$ (red) and $V = (1,1,1)/\sqrt{3}$ (blue), confirming robust $|0\rangle \to |1\rangle$ transfer.

Robust state-transfer



The sphere-averaged fidelity remains near unity for $|0\rangle \to |1\rangle$ transfer in the perturbative regime and degrades only weakly with α.

Multiple state pairs



The same robustness trend is observed across five representative state pairs on the Bloch sphere.

Gate Synthesis

Beyond robust state transfer, we seek, as a worked example, control pulses that robustly implement the NOT gate for any initial state.

Gate-level robustness

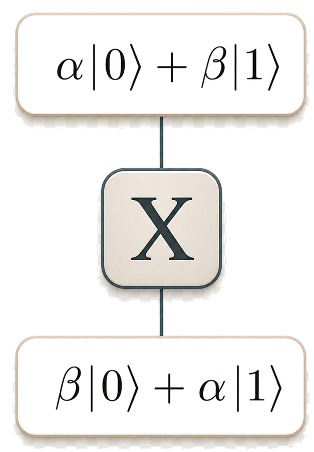
Analogously to the state-transfer case, we define E_k as:

$$E_k(t_f) = \int_0^{t_f} \langle\psi_0(t)|\sigma_k|\psi_0(t)\rangle\, dt, \qquad k \in \{x, y, z\}.$$

Unlike F_k, which involves both $|\psi_0\rangle$ and $|\psi_\perp\rangle$, the integrand here depends on $|\psi_0\rangle$ alone.

Robust synthesis requires both F_k and E_k to vanish simultaneously [1]. Thus, robust gate synthesis requires

$$F_k(t_f) = 0, \qquad E_k(t_f) = 0, \qquad k \in \{x, y, z\}.$$



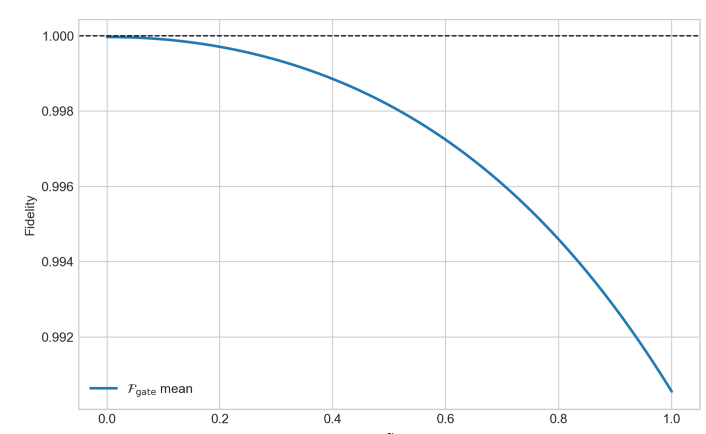
Evaluation metric

For each α, performance is measured by the gate fidelity:

$$\mathcal{F}_{\text{gate}} = \frac{1}{4}\left|\mathrm{Tr}\Big(U_{\text{NOT}}^{\dagger} U(\alpha, \hat{V})\Big)\right|^2$$

The displayed curve shows the mean gate fidelity over sampled perturbation directions $\hat{V}$. U_{NOT} is the ideal NOT unitary and $U(\alpha, \hat{V})$ is the propagator under $H_0(t) + \alpha(\hat{V} \cdot \vec{\sigma})$.

Robust gate fidelity



Mean gate fidelity remains close to unity across $\alpha \in [0, 1]$.

Future Directions

- Energy-efficient robust pulses
- Higher-order robustness
- Directional-to-universal robustness tradeoffs
- Dynamical-noise-resilient pulses

Want to connect?

Scan the QR code.

References & Acknowledgements

References

[1] O. Fresse-Colson, S. Guérin, X. Chen, and D. Sugny. *Application of the Pontryagin maximum principle to the robust time-optimal control of two-level quantum systems.* Phys. Rev. A **112**, 022618 (2025).

Acknowledgement

Funding from the French National Research Agency (ANR) under grant ANR-23-CPJ1-0030-01.

Université Bourgogne Europe

Laboratoire Interdisciplinaire Carnot de Bourgogne icb